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Mathematics · Module One

Numbers and Functions

Worked Solutions & Discussion

*Full solutions to every “Problem to work,”
discussion for the “Things worth thinking about,”
and what each “Teach it” task is really after.*

A companion to the textbook. Wrestle with every problem yourself first —
the answer is worth far less than the struggle that earns it.

How to use this

This booklet answers the exercises in *Numbers and Functions*, chapter by chapter.

- **Things worth thinking about** are open questions — several have no single “right” answer. What follows each is a *discussion*: the idea the question is fishing for, and where it leads. Yours may go somewhere different and still be excellent.
- **Problems to work** get *full solutions* — the reasoning, not just the final number. Where a problem has a definite answer, it’s shown in **bold**.
- **Teach it** problems can’t be “answered” on paper — their prize is what happens in your head while you explain out loud. For those, this booklet lists the key points a strong explanation lands.
- **[challenge]** marks the harder problems. If one didn’t yield at first, that’s the point of the mark — not a sign you did anything wrong.

One honest request: *try first*. Reading a solution feels like understanding, but the understanding only sticks if your own struggle came before it.

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Chapter 1 · Counting Isn't Enough

Things worth thinking about

1. **Why were $+$ and $\times$ fine, but $-$ and $\div$ broke?** Because adding or multiplying two counting numbers always lands you on another counting number — you only ever move “outward” into numbers you already have. Subtraction and division are the *inverse* (undo) operations, and undoing can demand an answer that lies *below* where you started (past zero) or *between* the whole numbers. The operations aren’t “bad”; it’s that the inverse of a building-up operation naturally reaches for answers outside the set you began with.
2. **What could $\frac{1}{0}$ be?** Suppose $\frac{1}{0} = n$. By the meaning of division that says $1 = 0 \times n$. But $0 \times n = 0$ for *every* number n , so this demands $1 = 0$ — impossible. No value of n works, so $\frac{1}{0}$ isn’t any number at all. (And $\frac{0}{0} = n$ would say $0 = 0 \times n$, true for *every* n — not a single answer but all of them at once.) Division by zero is banned because it is genuinely impossible or hopelessly ambiguous, not by decree.
3. **Are there “more” fractions than integers?** The surprise: *no* — *they are the same size of infinity*. Even though fractions are packed densely, you can arrange all of them in a grid and snake through it, pairing each fraction with a counting number $1, 2, 3, \dots$, so none is left out. (The *irrationals*, by contrast, really are a strictly bigger infinity — Chapter 2’s hint, and a later book’s story.)
4. **An operation even fractions can’t handle?** Taking a square root: no fraction, times itself, gives 2. That forces the irrational numbers (Chapter 2). Later, asking for a number whose square is -1 forces the complex numbers (Chapter 8).

Problems to work

Problem 1 — Teach it (the invention pattern). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* (i) An operation broke: $3 - 5$ has no counting-number answer, so we *invented* -2 . (ii) It happened again: $12 \div 5$ has no integer answer, so we invented $\frac{12}{5}$. (iii) The one sentence that is the whole point — “*an operation broke, so we built the number that fixes it.*” If your listener can say that back, they have it.

Problem 2 — Sort the zoo. $7 \in \mathbb{N}$ (counting); $-4 \in \mathbb{Z}$ (integer); $\frac{3}{8} \in \mathbb{Q}$

(rational); $0 \in \mathbb{Z}$ (an integer, but not a counting number here); $-\frac{10}{2} = -5 \in \mathbb{Z}$ (the sneaky one — it simplifies to an integer); $\frac{9}{3} = 3 \in \mathbb{N}$ (also sneaky — it's really 3).

Problem 3 — Squeeze one in. The average always works; here are easy choices. (a) between $\frac{1}{4}$ and $\frac{1}{2}$: their average is $\frac{3}{8}$. (b) between $\frac{2}{5}$ and $\frac{3}{5}$: $\frac{1}{2}$ (it's $\frac{2.5}{5}$). (c) between $\frac{3}{7}$ and $\frac{4}{7}$: write them as $\frac{6}{14}$ and $\frac{8}{14}$ — right between sits $\frac{7}{14} = \frac{1}{2}$.

Problem 4 — The divide-by-zero trap. For *any* n , $0 \times n = 0$. So $\frac{1}{0} = n$ would force $1 = 0$, which is false — hence no n can work, and dividing by zero is impossible, not merely forbidden.

Problem 5 — [challenge] Always room for one more. Let the two fractions be $\frac{a}{b}$ and $\frac{c}{d}$ (integers, $b, d \neq 0$). Their average is

$$m = \frac{1}{2}\left(\frac{a}{b} + \frac{c}{d}\right) = \frac{ad + bc}{2bd}.$$

(i) **It's a fraction:** $ad + bc$ and $2bd$ are integers and $2bd \neq 0$, so m is a ratio of integers — rational. (ii) **It's strictly between:** the average of two *different* numbers always lies strictly between them, so if $\frac{a}{b} < \frac{c}{d}$ then $\frac{a}{b} < m < \frac{c}{d}$. Since this works for *any* two distinct fractions, you can now do it again between $\frac{a}{b}$ and m , and again, forever: every gap, however small, holds infinitely many more fractions.

Problem 6 — [challenge] Is there a smallest positive fraction? No. Smaller than $\frac{1}{1,000,000,000}$ is, for instance, $\frac{1}{1,000,000,001}$ — or just halve it to $\frac{1}{2,000,000,000}$. In general, given *any* positive fraction $\frac{p}{q}$, the fraction $\frac{p}{q+1}$ (bigger bottom) or $\frac{p}{2q}$ is positive and smaller. There is always a smaller one, so no smallest exists.

Problem 7 — [challenge] The next wall. $\sqrt{2} = 1.4142135\dots$ Squaring the eight-digit value 1.4142135 gives 1.99999982..., *not* exactly 2. No terminating decimal you write down will square to exactly 2 — because a terminating decimal is a fraction, and (as Chapter 2 proves) *no* fraction squares to 2. The square-root operation has broken free of the fractions, and that is precisely the wall Chapter 2 climbs.

Chapter 2 · The Hole in the Number Line

Things worth thinking about

1. **Why does an even square force an even number?** An odd number is $2k + 1$, and $(2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ — odd. So odd $\times$ odd is always odd; turned around, if a square is *even*, the number can't have been odd, so it was even. (Check: $3^2=9$, $5^2=25$, $7^2=49$ — all odd.)
2. **Does the proof work for $\sqrt{4}$?** Run it: $\sqrt{4} = \frac{a}{b}$ in lowest terms gives $a^2 = 4b^2$, so a is even, $a = 2k$, giving $4b^2 = 4k^2$, i.e. $b^2 = k^2$ — and that's *no contradiction* ($a = 2$, $b = 1$ works fine). The argument breaks exactly because 4 is a perfect square: $\sqrt{4} = 2$, an honest rational. The “break” is the proof telling you the truth.
3. **Is 0.1010010001... rational? Irrational.** It never stops, and the growing gaps mean it never settles into a fixed repeating block — and only “stops or repeats” decimals are fractions.
4. **Same size of infinity, or bigger? Some infinities are bigger.** The irrationals (and the reals) form a strictly larger infinity than the rationals — the rationals can be listed off against $1, 2, 3, \dots$, but the reals provably cannot.
5. **Binary.** $10000_2 = 16$ (the lone 1 sits in the sixteens place: $1, 2, 4, 8, 16$). And $10 = 8 + 2 = 1010_2$.
6. **Are computers bad at math?** No — it's the price of *finite* storage. In binary, 0.1 and 0.2 are unending repeating fractions, so the machine rounds each to the nearest value it can hold; the two roundings don't quite cancel, leaving 0.3000...04. The arithmetic is exact *for the rounded inputs* — the speck of error is storage, not stupidity.

Problems to work

Problem 1 — Turn the crank ($\sqrt{3}$ is irrational). Suppose $\sqrt{3} = \frac{a}{b}$ in lowest terms (so a, b are not both multiples of 3). Then $3 = \frac{a^2}{b^2}$, i.e. $3b^2 = a^2$, so 3 divides a^2 . Since 3 is prime, 3 divides a ; write $a = 3k$. Then $3b^2 = 9k^2$, so $b^2 = 3k^2$, giving $3 \mid b^2$ and hence $3 \mid b$. Now 3 divides *both* a and b — contradicting “lowest terms.” So $\sqrt{3}$ is irrational. (*The one fact you need is “ $3 \mid a^2 \Rightarrow 3 \mid a$,” true because 3 is prime — the exact analogue of the even/odd fact.*)

Problem 2 — Teach it ($\sqrt{2}$ is irrational). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* (i) Assume

$\sqrt{2} = \frac{a}{b}$ in lowest terms, so a, b are not both even. (ii) From $2b^2 = a^2$, a^2 is even, so a is even; writing $a = 2k$ gives $b^2 = 2k^2$, so b is even too. (iii) “Both even” contradicts “lowest terms” — that contradiction is what kills the assumption, so no such fraction exists.

Problem 3 — Stretch ($\sqrt[3]{2}$ is irrational). Suppose $\sqrt[3]{2} = \frac{a}{b}$ in lowest terms. Cube: $2 = \frac{a^3}{b^3}$, so $2b^3 = a^3$, giving $2 \mid a^3$ and therefore (free fact) $2 \mid a$. Write $a = 2k$: then $2b^3 = 8k^3$, so $b^3 = 4k^3 = 2(2k^3)$, giving $2 \mid b^3$ and so $2 \mid b$. Both a and b are even — contradiction. Hence $\sqrt[3]{2}$ is irrational. *Same engine as before, one gear changed: squaring became cubing.*

Problem 4 — Teach it (the computer’s sketch). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* the machine isn’t broken; it stores every number at the nearest dot on a *finite* grid. 0.1 and 0.2 land a hair off their true spots (they’re unending in binary), so their sum lands a hair off 0.3. The keeper sentence: “*the computer holds a very good sketch of each number, never the number itself.*”

Problem 5 — Repeating decimal back to a fraction. Let $x = 0.\overline{27}$. Then $100x = 27.\overline{27}$. Subtract: $100x - x = 27$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$.

Problem 6 — [challenge] Rational, or not? (a) $0.454545\dots$ **rational** (it repeats; in fact $\frac{45}{99} = \frac{5}{11}$). (b) $\sqrt{9} = 3$ — **rational**. (c) $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$ — **irrational** (a nonzero rational times an irrational is irrational). (d) $\pi - \pi = 0$ — **rational**. (So “contains a root/ π ” $\neq$ “irrational.”)

Problem 7 — [challenge] A sum that hides. Yes, it can happen. $\sqrt{2} + (-\sqrt{2}) = 0$, and $\sqrt{2} + (4 - \sqrt{2}) = 4$ — both sums are rational even though every piece ($\sqrt{2}$, $-\sqrt{2}$, $4 - \sqrt{2}$) is irrational. So irrational + irrational is *sometimes* rational (and usually isn’t).

Problem 8 — [challenge] Irrational to an irrational power. Let $N = \sqrt{2}^{\sqrt{2}}$. *Case 1:* if N is rational, then $(\sqrt{2}, \sqrt{2})$ is already an irrational-base, irrational-power pair giving a rational — done. *Case 2:* if N is irrational, then

$$N^{\sqrt{2}} = \left(\sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2,$$

rational — so $(N, \sqrt{2})$ is the pair. Either way such a pair exists. The beautiful part: we never had to find out *which* case is true.

Chapter 3 · Functions as Machines

Things worth thinking about

1. **Two inputs, one output — and one-to-one machines.** Many machines send different inputs to the same output: $f(x) = |x|$ (both $\pm 3 \mapsto 3$), $f(x) = x^4$, or cosine. A machine that *never* does this — every output from exactly one input — is one that only ever climbs (or only ever falls): $f(x) = x + 1$, $f(x) = x^3$, $f(x) = 2x$. Those are the ones with clean undo buttons.
2. **When does order not matter?** When the two machines are the same *kind*. Two “add” machines: $f(x) = x + 2$, $g(x) = x + 3$ give $f(g(x)) = g(f(x)) = x + 5$. Two “multiply” machines: $f(x) = 2x$, $g(x) = 3x$ both give $6x$. What’s special is that adding commutes with adding, and scaling with scaling — mix an add with a multiply and the order bites again.
3. **Top half of a circle.** The full circle fails the vertical-line test because one x gives two heights ($\pm$). Keep only the *upper* arc and each x gives a single height — it passes. Lesson: a “non-function” can be sliced into pieces that *are* functions (here, upper and lower arcs) — exactly how $\sqrt{}$ is carved out of x^2 .
4. **Narrowing the domain.** Yes. If we only ever feed squaring numbers from $[0, \infty)$, it never sees both 3 and -3 , so nothing collides and it can be undone — the undo is $\sqrt{}$. Shrinking the domain to kill collisions is the trick that rescues an inverse (used constantly for trig later).

Problems to work

Problem 1 — Read the machine. (a) $f(x) = x + 7$: nothing jams. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$. (b) $g(x) = \frac{1}{x-2}$: jams when $x - 2 = 0$, i.e. $x = 2$. Domain $(-\infty, 2) \cup (2, \infty)$; it can output any value except 0, so range $(-\infty, 0) \cup (0, \infty)$. (c) $h(x) = \sqrt{x-4}$: needs $x - 4 \geq 0$, i.e. $x \geq 4$. Domain $[4, \infty)$ (*square bracket* — $x = 4$ gives $\sqrt{0} = 0$, which is fine); range $[0, \infty)$.

Problem 2 — Chain them, both ways. $f(x) = 2x$, $g(x) = x + 3$. (a) $f(g(5)) = f(8) = \mathbf{16}$. (b) $g(f(5)) = g(10) = \mathbf{13}$. (c) Adding 3 *first* means the doubling acts on it too ($16 = 2 \cdot 5 + 2 \cdot 3$), but doubling first and then adding 3 only adds it once. The order decides whether the +3 gets doubled.

Problem 3 — Teach it (the inverse idea). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* output 9 from a squaring machine could have come from 3 *or* -3 — the machine threw

away the sign, so it can't be run backwards. The feeling to pass on: *the ambiguity is the obstruction*; an undo exists only when no two inputs ever collide.

Problem 4 — Teach it (domain jams). *A teaching task — the goal is what happens in your head while you explain. A strong version hits: $g(x) = \frac{1}{x-2}$ happily eats 100 or -7 , but at $x = 2$ the bottom is 0 and division by zero is impossible — the calculator complains for the same reason. The idea to land: a domain isn't an arbitrary rule, it's just "the inputs that don't break the machine."*

Problem 5 — Evaluate carefully. $f(x) = x^2 - 3x + 1$. $f(0) = \mathbf{1}$; $f(2) = 4 - 6 + 1 = -1$; $f(-1) = 1 + 3 + 1 = \mathbf{5}$ (watch: $-3(-1) = +3$); $f(\frac{1}{2}) = \frac{1}{4} - \frac{3}{2} + 1 = -\frac{1}{4}$.

Problem 6 — Match the graph to its rule. (a) vertical line — **not** a function (one x , infinitely many y). (b) horizontal line — **is** a function. (c) "V" — **is** (like $|x|$). (d) full circle — **not** ($x = 0$ gives two y). (e) sideways "S" — **not** (a vertical line cuts it more than once). For each "no," a single vertical line (e.g. through the middle) meets the curve twice.

Problem 7 — [challenge] Build the inverse. $f(x) = 2x + 6$ doubles then adds 6; undo in reverse — subtract 6, then halve:

$$f^{-1}(x) = \frac{x - 6}{2}.$$

Test: $f(5) = 16$, and $f^{-1}(16) = \frac{16-6}{2} = 5$. ✓

Problem 8 — [challenge] Narrowing the domain rescues an inverse. (a) On $[0, \infty)$ the squaring machine never sees two inputs with the same square (no negatives to collide with the positives), so it's now one-to-one and *can* be undone. (b) The undo machine is $\sqrt{x}$. (c) That's exactly why $\sqrt{}$ is *defined* to return only the non-negative root — so it is a genuine, single-valued inverse of squaring-on- $[0, \infty)$.

Problem 9 — [challenge] Compose three. $a(x) = x + 1$, $b(x) = 2x$, $c(x) = x^2$. $a(b(c(2)))$: $c(2) = 4 \rightarrow b = 8 \rightarrow a = \mathbf{9}$. $c(b(a(2)))$: $a(2) = 3 \rightarrow b = 6 \rightarrow c = \mathbf{36}$ — different. And $a(b(c(x))) = a(b(x^2)) = a(2x^2) = \mathbf{2x^2 + 1}$.

Chapter 4 · Polynomials, Factoring, the Quadratic Formula

Things worth thinking about

1. **The constant 7.** Its graph is the horizontal line $y = 7$. It has **no roots** — 7 is never 0. (A nonzero constant never crosses the axis.)
2. **Two / one / zero roots.** A parabola opening upward has two roots when its vertex sits *below* the axis (it cuts through twice), one (repeated) root when the vertex sits *on* the axis (it just kisses), and no real roots when the vertex floats *above* the axis (it never reaches down). It's all about where the bottom of the U sits relative to the axis.
3. **Why count roots without finding them?** To know *whether* real answers exist before committing to the work: does a thrown ball ever reach a given height? does a design have a real solution? Sometimes you only need “can this happen at all?” — and the discriminant answers that in one cheap step.
4. **Roots a curve never touches.** If the discriminant is negative, the roots are *complex* — they live off the real number line, so the real-plane graph shows no crossing. They're real solutions, just not on the real axis (Chapter 8 builds the plane they live in).
5. **“Not found yet” vs “proven impossible.”** The first is ignorance; the second is knowledge — a proof that *no* method can ever succeed (Abel for the quintic). Other proven impossibilities: trisecting a general angle or “squaring the circle” with straightedge and compass; reaching certain positions in a sliding puzzle. Proving impossibility is a deeper act than failing to find.

Problems to work

Problem 1 — Factor and read the roots. (a) $x^2 - 7x + 12 = (x - 3)(x - 4)$, roots **3, 4**. (b) $x^2 + x - 6 = (x + 3)(x - 2)$, roots **-3, 2**. (c) $x^2 - 9 = (x - 3)(x + 3)$, roots **± 3** (the two numbers multiply to -9 and add to 0: 3 and -3).

Problem 2 — Complete the square. $x^2 + 8x + 7 = 0 \Rightarrow x^2 + 8x = -7$. Half of 8 is 4; $4^2 = 16$. Add 16 to both sides: $x^2 + 8x + 16 = 9$, i.e. $(x + 4)^2 = 9$, so $x + 4 = \pm 3$, giving $x = -1$ or $x = -7$. Check: $(x + 1)(x + 7) = x^2 + 8x + 7$. ✓

Problem 3 — Turn the crank. $x^2 - 4x + 1 = 0$: $a = 1, b = -4, c = 1$.

Discriminant $b^2 - 4ac = 16 - 4 = 12$.

$$x = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}.$$

(The leftover $\sqrt{3}$ is why no nice factoring existed.)

Problem 4 — Read the fork. (a) $x^2 - 6x + 9$: $36 - 36 = 0$ — **one repeated** real root. (b) $x^2 + 2x + 5$: $4 - 20 = -16 < 0$ — **no real** roots (pocket it for Chapter 8). (c) $2x^2 + 3x - 1$: $9 - 4(2)(-1) = 17 > 0$ — **two real** roots.

Problem 5 — Teach it (the zero-product idea). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:*

if $(x - 2)(x - 3) = 0$, then one factor must be 0 (the only way a product is zero), so $x = 2$ or $x = 3$. But $(x - 2)(x - 3) = 12$ does *not* let you set each piece to 12 — 12 can be made from many pairs ($12 = 3 \times 4 = 6 \times 2 = \dots$), and only *zero* has the “one factor must be it” property. You must expand: $x^2 - 5x + 6 = 12 \Rightarrow x^2 - 5x - 6 = 0 \Rightarrow (x - 6)(x + 1) = 0 \Rightarrow x = 6, -1$.

Problem 6 — Verify a toolkit identity. $(a - b)(a^2 + ab + b^2) = a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3$. The middle four terms cancel in pairs (a^2b with $-a^2b$, ab^2 with $-ab^2$), leaving $a^3 - b^3$. ✓

Problem 7 — Use the toolkit. (a) $x^2 - 49 = (x - 7)(x + 7)$ — *difference of squares*. (b) $x^2 + 10x + 25 = (x + 5)^2$ — *perfect-square trinomial*. (c) $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$ — *difference of cubes* ($a = x$, $b = 2$).

Problem 8 — Extend the triangle. Below 1, 4, 6, 4, 1: next row 1, 5, 10, 10, 5, 1; then 1, 6, 15, 20, 15, 6, 1. The fifth-power row gives

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5.$$

Problem 9 — Teach it (frozen thought). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* the quadratic formula isn’t a magic incantation — it is completing-the-square done *once*, with letters instead of numbers, so nobody has to repeat the four steps. Show $x^2 + 6x + 5 = 0$ by hand, then point at $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ as the *same moves* on a, b, c . The keeper: *a formula is frozen reasoning — forget it and you can thaw it back out.*

Problem 10 — [challenge] A disguised difference of squares. $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$. Over the reals $x^2 + 4$ won’t factor further, so the complete real factorization is $(\mathbf{x} - 2)(\mathbf{x} + 2)(\mathbf{x}^2 + 4)$. (That stubborn $x^2 + 4$ factors only once we have i — Chapter 8.)

Problem 11 — [challenge] Build a quadratic to order. Roots 5 and -2 : $(x - 5)(x + 2) = \mathbf{x^2 - 3x - 10}$. Roots 3 and 3: $(x - 3)^2 = \mathbf{x^2 - 6x + 9}$, with discriminant $36 - 36 = 0$ — a repeated root, as it should be.

Problem 12 — [challenge] Sum and product of roots. $(x - r)(x - s) = x^2 - (r + s)x + rs$. Matching $x^2 + bx + c$ gives $r + s = -b$ and $rs = c$. So two numbers adding to 7 and multiplying to 10 are **2** and **5**, hence $x^2 - 7x + 10 = (x - 2)(x - 5)$.

Problem 13 — [challenge] The geometric-sum payoff. With $x = 2$ and last term $1024 = 2^{10}$ (so $n = 11$ terms),

$$1 + 2 + 4 + \cdots + 1024 = \frac{2^{11} - 1}{2 - 1} = 2^{11} - 1 = \mathbf{2047}.$$

The full chessboard (2^0 through 2^{63}) totals $2^{64} - 1 = 18,446,744,073,709,551,615$ grains — more rice than has ever been grown.

Chapter 5 · Exponentials and Logarithms

Things worth thinking about

1. **What is $2^{1/3}$?** The number that, multiplied by itself *three* times, gives 2 — the cube root, $\sqrt[3]{2}$. (And you proved in Chapter 2 it's irrational.)
2. **Why can 2^x never be 0?** A power of 2 is built only by multiplying and dividing 2's, and that can never land on exactly nothing: $2^{-n} = \frac{1}{2^n}$ is a positive fraction that shrinks toward 0 but stays positive forever. Positive times positive is positive — zero is unreachable.
3. **Why was “ $\times \rightarrow +$ ” worth gold?** Before calculators, multiplying two big numbers by hand was slow and error-prone. With logarithms you could *look up, add, look back* — turning a hard multiplication into an easy addition. For a navigator or astronomer multiplying huge figures nightly, that was the difference between hours and minutes (and far fewer mistakes).
4. **What grows at a rate equal to its size?** Money under compound interest (more money $\rightarrow$ more interest $\rightarrow$ more money); a population (more individuals $\rightarrow$ more births); bacteria in a dish; a nuclear chain reaction; a rumor in its early spread. Anything where *how fast it grows is proportional to how much there already is* — which is the defining trait of e^x .

Problems to work

Problem 1 — Forced meanings. (a) $2^3 \cdot 2^0 = 2^3$ forces $2^0 = 1$. (b) $2^2 \cdot 2^{-2} = 2^0 = 1$ forces $2^{-2} = \frac{1}{4}$. (c) $2^{1/2} \cdot 2^{1/2} = 2^1 = 2$ forces $2^{1/2} = \sqrt{2}$.

Problem 2 — Simplify with the exponent laws. (a) $a^4 \cdot a^7 = a^{11}$ (product). (b) $\frac{a^9}{a^5} = a^4$ (quotient). (c) $(a^3)^4 = a^{12}$ (power of a power). (d) $\frac{a^2 a^6}{a^3} = a^{8-3} = a^5$ (product, then quotient). (e) $a^{-3} \cdot a^5 = a^2$ (product). (f) $(2a^2)^3 = 2^3 a^6 = 8a^6$ (power of a product).

Problem 3 — Translate. (a) $3^4 = 81 \iff \log_3 81 = 4$. (b) $\log_{10} 1000 = 3 \iff 10^3 = 1000$. (c) $\log_2 \frac{1}{4} = -2 \iff 2^{-2} = \frac{1}{4}$.

Problem 4 — Read the log as a question. (a) $\log_2 16 = 4$. (b) $\log_3 9 = 2$. (c) $\log_5 1 = 0$. (d) $\log_2 \frac{1}{8} = -3$.

Problem 5 — Use the log laws. (a) $\log_a(7x) = \log_a 7 + \log_a x$. (b) $\log_a \frac{x}{5} = \log_a x - \log_a 5$. (c) $\log_a(x^3) = 3 \log_a x$. (d) $\log_a x + \log_a y - \log_a z = \log_a \frac{xy}{z}$. (e)

$$2^x = 50 \Rightarrow x = \log_2 50 = \frac{\ln 50}{\ln 2} \approx \mathbf{5.64}.$$

Problem 6 — Teach it (x^2 vs 2^x). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* in x^2 the variable is the *base* (a fixed power of a changing number); in 2^x it's the *exponent* (a fixed number raised to a changing power). At $x = 2$ they tie ($4 = 4$); at $x = 10$, $10^2 = 100$ but $2^{10} = 1024$. They actually cross at $x = 2$ and $x = 4$, after which 2^x pulls ahead forever. The keeper: *the variable being upstairs is what makes the exponential eventually outrun any power.*

Problem 7 — Compounding, and its ceiling. (a) Balance after x years: $B(x) = 1000(1.05)^x$. (b) After 1 yr, \$1050; after 2 yr, $1000(1.1025) = \$1102.50$ — the second year's gain (\$52.50) beats the first's (\$50): interest on interest. (c) $1000(1.05)^{300} \approx \2.3 billion. No real deposit does this because the model is the *unconstrained early phase*; reality is an S-curve. The ceiling is the finite size of the economy (and rate changes, bank failures, inflation, wars) — the constraint that always, eventually, bites.

Problem 8 — Teach it (the sigmoid decision). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* feed $\sigma(x) = \frac{1}{1+e^{-x}}$ a big positive number and it answers near 1 (“yes”); a big negative number, near 0 (“no”); and 0 gives exactly $\frac{1}{2}$ (“unsure”). The punchline: this one curve is how an artificial neuron decides *and* the shape an epidemic follows as it levels off — the same S doing both jobs.

Problem 9 — [challenge] Half-life, backwards. $A(t) = 100 \left(\frac{1}{2}\right)^{t/10}$. (a) After 30 yr: $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, so $\frac{100}{8} = \mathbf{12.5}$ g. (b) $25 = 100 \left(\frac{1}{2}\right)^{t/10} \Rightarrow \left(\frac{1}{2}\right)^{t/10} = \frac{1}{4} = \left(\frac{1}{2}\right)^2 \Rightarrow t/10 = 2 \Rightarrow t = \mathbf{20}$ years. (c) $10 = 100 \left(\frac{1}{2}\right)^{t/10} \Rightarrow \left(\frac{1}{2}\right)^{t/10} = \frac{1}{10} \Rightarrow t = 10 \log_2 10 = \frac{10 \ln 10}{\ln 2} \approx \mathbf{33.2}$ years.

Problem 10 — [challenge] The log scale's superpower. (a) Magnitude 7 vs 5 is 2 steps, so $32^2 = \mathbf{1024}$ times the energy. (b) Because $\log(xy) = \log x + \log y$: multiplying energy by 32 *adds* $\log 32$ to its logarithm, so “magnitude” (a logarithm of energy) just goes up by 1 per step — billion-fold energy ranges become a tidy 1–10 scale. (c) A plain-joule scale would run from the tiny to the astronomically huge; the numbers for a “4” and an “8” would differ by a factor of a million, making them impossible to compare at a glance. The logarithm compresses that into something human-sized.

Problem 11 — [challenge] Stacked exponents. $2^{(3^2)} = 2^9 = \mathbf{512}$, while

$(2^3)^2 = 8^2 = \mathbf{64}$. So $2^{(3^2)}$ is far bigger. Towers are read *top-down*: $(a^m)^n = a^{mn}$ (here $2^6 = 64$) is *not* the same as $a^{(m^n)}$ (here 2^9).

Chapter 6 · Trigonometric Functions

Things worth thinking about

1. **Reading the circle.** $\sin \theta = 1$ when the walker is at the *top* ($\theta = \frac{\pi}{2}$); $\cos \theta = -1$ when the walker is at the *left* ($\theta = \pi$). They cannot both be 1: then $\cos^2 \theta + \sin^2 \theta = 1 + 1 = 2 \neq 1$, impossible.
2. **A wave with period 1.** Use $\sin(2\pi\theta)$: as θ runs from 0 to 1, the input $2\pi\theta$ runs through a full 2π — one complete cycle. Multiplying the input speeds the walker up; the factor 2π packs a whole loop into one unit.
3. **Why tangent blows up.** $\tan \theta$ is the slope of the line from the center to the walker. When $\cos \theta = 0$ (walker straight up or down) that line is *vertical* — all rise, no run — and slope is $\text{rise} \div \text{run}$, so you'd be dividing by zero. The line standing straight up is the geometric face of the infinite slope.
4. **e and circles (a guess).** For the growth machine and the going-around machine to coincide, the new number i would have to convert *outward* push into *sideways* push — i.e. multiplying by i is a quarter-turn. Then feeding e an imaginary input makes it curve around instead of shooting off, tracing a circle. (That's exactly $e^{i\theta}$, Chapter 8.)

Problems to work

Problem 1 — Read the circle. (a) $\theta = 0$: $\cos = 1, \sin = 0$ (right). (b) $\frac{\pi}{2}$: $\cos = 0, \sin = 1$ (top). (c) π : $\cos = -1, \sin = 0$ (left). (d) $\frac{3\pi}{2}$: $\cos = 0, \sin = -1$ (bottom). (e) 2π : $\cos = 1, \sin = 0$ — *identical to* $\theta = 0$; one full lap returns the walker home.

Problem 2 — Use the Pythagorean identity. $\cos \theta = \frac{3}{5} \Rightarrow \sin^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$, so $\sin \theta = \pm \frac{4}{5}$. Two values because the walker could be above the axis ($+\frac{4}{5}$) or below it ($-\frac{4}{5}$) at that same across-position.

Problem 3 — Derive $\sin 2\theta$. $\sin(\theta + \theta) = \sin \theta \cos \theta + \cos \theta \sin \theta = 2 \sin \theta \cos \theta$.

Problem 4 — $\cos 2\theta$, three faces. $\cos(\theta + \theta) = \cos^2 \theta - \sin^2 \theta$. Using $\sin^2 \theta = 1 - \cos^2 \theta$ gives $\cos 2\theta = 2 \cos^2 \theta - 1$; using $\cos^2 \theta = 1 - \sin^2 \theta$ gives $\cos 2\theta = 1 - 2 \sin^2 \theta$. Three costumes for one fact.

Problem 5 — Special angles from the triangles. (a) Half a unit square: legs 1, 1, hypotenuse $\sqrt{2}$; scaled to hypotenuse 1, $\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$. (b) Half

an equilateral triangle of side 2: base 1, height $\sqrt{3}$, hypotenuse 2; scaled to hypotenuse 1, $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$, $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\sin \frac{\pi}{6} = \frac{1}{2}$, $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$.

Problem 6 — Reductions off the walker. (a) $\sin \frac{2\pi}{3} = \sin(\pi - \frac{\pi}{3}) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$ (same height, mirrored left). (b) $\cos \frac{2\pi}{3} = -\cos \frac{\pi}{3} = -\frac{1}{2}$ (opposite across-position). (c) $\cos \frac{\pi}{3} = \sin(\frac{\pi}{2} - \frac{\pi}{3}) = \sin \frac{\pi}{6} = \frac{1}{2}$ (the complement swap).

Problem 7 — Derive $\tan 2\theta$. Divide top and bottom of $\frac{2 \sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta}$ by $\cos^2 \theta$. Top: $\frac{2 \sin \theta \cos \theta}{\cos^2 \theta} = 2 \tan \theta$. Bottom: $\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} = 1 - \tan^2 \theta$. Hence $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$.

Problem 8 — Teach it (sin is bounded). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* the walker is pinned to a circle of radius 1, so its height can never exceed 1 or drop below -1 — no formula needed, just the picture. The keeper: *a bounded machine is a different animal from the exponential that ran off to infinity.*

Problem 9 — Teach it (why cos is “co-sine”). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* reflect the walker at angle θ across the 45° diagonal: “how high” and “how far across” swap, so $\sin(\frac{\pi}{2} - \theta) = \cos \theta$. The keeper: *cosine is just sine measured from the perpendicular direction* — which is why the two waves are the same shape, a quarter-turn apart.

Problem 10 — State the domains. (a) $\tan \theta = \frac{\sin}{\cos}$ jams where $\cos \theta = 0$: the odd multiples of $\frac{\pi}{2}$ ($\frac{\pi}{2}, \frac{3\pi}{2}, \dots$). (b) $\cot \theta = \frac{\cos}{\sin}$ jams where $\sin \theta = 0$: the multiples of π ($0, \pi, 2\pi, \dots$). (c) $\csc \theta = \frac{1}{\sin}$ jams where $\sin \theta = 0$ — same forbidden angles as $\cot \theta$. (d) $\sin \theta$ and $\cos \theta$ are the walker’s coordinates directly — no denominator, so nothing to divide by zero; defined for *every* angle.

Problem 11 — [challenge] Build a wave to order. Amplitude 3 scales the height; period π needs an inside factor b with $\frac{2\pi}{b} = \pi$, i.e. $b = 2$. So $y = 3 \sin(2\theta)$ — it swings between -3 and 3 and repeats every π .

Problem 12 — [challenge] A surprising sum. $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}} = \frac{\sqrt{6} + \sqrt{2}}{4} \approx 0.966$.

Problem 13 — [challenge] The identity behind a sunset. Start from $\cos 2\theta = 1 - 2 \sin^2 \theta$. Solve for $\sin^2 \theta$: $2 \sin^2 \theta = 1 - \cos 2\theta$, so $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$. (This is the workhorse that lets you integrate $\sin^2$ later.)

Chapter 7 · Functions as Objects

Things worth thinking about

1. **The left-shift check.** $f(x+3) = (x+3)^2$ bottoms out where the inside is zero: $x+3=0$, i.e. $x=-3$ — to the *left* of 0. So adding inside shifts the graph left, exactly as claimed.
2. $f(x) = x^2 + x$ — **even, odd, or neither?** $f(-x) = x^2 - x$. That's not equal to $f(x) = x^2 + x$ (not even), nor to $-f(x) = -x^2 - x$ (not odd). So **neither**.
3. **Transformation as composition.** $2f(x)$ is f followed by a “multiply-by-2” machine — it acts *after* f (on the output). $f(2x)$ is a “multiply-by-2” machine followed by f — it acts *before* f (on the input). Same little machine, opposite side.
4. **A guess about i and circles.** A number whose square is negative behaves like a *rotation*: do it twice (square it) and you've turned a half turn, which points you backward — i.e. multiplies by -1 . So multiplying by it is a quarter turn, and feeding it to e curls growth into a circle. (Any sensible guess in this spirit is the right instinct.)

Problems to work

Problem 1 — Read the transformations (base x^2). (a) $f(x) - 5$: down 5. (b) $f(x - 2)$: right 2 (inside, so it moves the opposite way to the sign — “ -2 inside” means right). (c) $3f(x)$: vertical stretch by 3 (three times as tall). (d) $f(-x)$: reflect across the vertical axis — *invisible* for x^2 , because the parabola is even (its mirror image is itself). (e) $f(x - 1) + 4$: right 1 and up 4 (vertex moves to $(1, 4)$).

Problem 2 — Build a wave to order. $2\sin(2x) + 3$. The “2” in front makes it twice as tall; the “2” inside makes it repeat twice as fast (period π); the “+3” lifts it up by 3.

Problem 3 — Even, odd, or neither. (a) x^4 : $f(-x) = x^4 = f(x)$ — **even**. (b) $x^3 - x$: $f(-x) = -x^3 + x = -(x^3 - x)$ — **odd**. (c) $x^2 + 1$: $f(-x) = x^2 + 1 = f(x)$ — **even**. (d) $x + 1$: $f(-x) = -x + 1$, which is neither $f(x)$ nor $-f(x) = -x - 1$ — **neither**.

Problem 4 — Iterate by hand. (a) $\frac{x}{2}$ from 32: 32, 16, 8, 4, 2, 1, ... — marches toward **0**. (b) $x + 3$ from 0: 0, 3, 6, 9, 12, 15, ... — **runs off** to infinity, never

settles. (c) $\frac{1}{2}(x + \frac{2}{x})$ from 1: $1 \rightarrow 1.5 \rightarrow 1.41\overline{6} \rightarrow 1.41421\dots$ — homing in on $\sqrt{2} = 1.41421\dots$ after just three steps.

Problem 5 — Watch chaos appear. (a) $r = 2$, start 0.3: $0.42, 0.4872, 0.4997, \dots \rightarrow 0.5$ — settles to a single value. (b) $r = 3.9$, start 0.3: the values **jump around** with no settling and no repeating block — chaos. (c) start 0.301 instead: for a few steps it shadows part (b), then **drifts completely apart** — the butterfly effect, by hand.

Problem 6 — Your first induction proof (odd numbers). *Base* $n = 1$: $1 = 1^2$. ✓ *Step*: assume $1 + 3 + \dots + (2k - 1) = k^2$. Add the next odd number $(2k + 1)$:

$$k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2,$$

exactly the formula at $n = k + 1$. So it holds for all n .

Problem 7 — Teach it (the left shift). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* whatever f used to do at $x = 3$, the machine $f(x + 3)$ now does at $x = 0$, because you fed it $0 + 3$. “Adding inside makes the behavior happen *earlier*,” so the graph slides *left*. The keeper is that example, not the rule “inside is backwards.”

Problem 8 — Teach it (the domino proof). *A teaching task — the goal is what happens in your head while you explain. A strong version hits:* knock the first domino (*base case*); guarantee each domino is close enough to tip the next (*inductive step*); conclude the whole infinite line falls — without checking every one. Tie it to $1 + 2 + \dots + n = \frac{n(n+1)}{2}$: $n = 1$ is the first tip, “if k then $k + 1$ ” is the spacing.

Problem 9 — [challenge] A second induction (doubling sum). *Base* $n = 1$: $1 = 2^1 - 1$. ✓ *Step*: assume $1 + 2 + \dots + 2^{k-1} = 2^k - 1$. Add 2^k :

$$(2^k - 1) + 2^k = 2 \cdot 2^k - 1 = 2^{k+1} - 1,$$

the formula at $k + 1$. Done.

Problem 10 — [challenge] Find the fixed point. (a) $f(x) = \frac{x}{2}$: $f(x) = x \Rightarrow \frac{x}{2} = x \Rightarrow x = 0$ — and 0 is exactly the attractor halving crawls toward. (b) $f(\sqrt{2}) = \frac{1}{2}(\sqrt{2} + \frac{2}{\sqrt{2}}) = \frac{1}{2}(\sqrt{2} + \sqrt{2}) = \sqrt{2}$ — a fixed point. (c) If an iteration settles to L , then L and $f(L)$ are the same value (the next term equals the current one), so $f(L) = L$: a landing spot must be a fixed point. That’s why these machines converge where they do.

Problem 11 — [challenge] Compose to undo. $f(x) = 3x - 6$. (a) Undo in reverse: add 6, then divide by 3: $g(x) = \frac{x+6}{3}$. (b) $f(g(x)) = 3 \cdot \frac{x+6}{3} - 6 = x$, and

$g(f(x)) = \frac{(3x-6)+6}{3} = x$. ✓ (c) Composing a machine with its inverse undoes every step in reverse, landing back where you started — the “do-nothing” (identity) machine.

Induction practice

Problem 12 — Even numbers. *Base* $n = 1$: $2 = 1 \cdot 2$. ✓ *Step*: assume $2 + 4 + \cdots + 2k = k(k+1)$. Add $2(k+1)$:

$$k(k+1) + 2(k+1) = (k+1)(k+2),$$

which is the formula at $k+1$. ✓

Problem 13 — [challenge] Divisibility. Claim: $3 \mid n^3 - n$. *Base* $n = 1$: $1 - 1 = 0$, divisible by 3. *Step*: assume $k^3 - k = 3m$. Then

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1 = (k^3 - k) + 3k^2 + 3k = 3m + 3(k^2 + k) = 3(m + k^2 + k).$$

A multiple of 3. ✓ (No sum in sight — the work is all in the step.)

Problem 14 — [challenge] An inequality. Claim: $n < 2^n$ for $n \geq 1$. *Base* $n = 1$: $1 < 2$. *Step*: assume $k < 2^k$. Since $k, 2^k$ are whole numbers, $k+1 \leq 2^k$; and $2^k < 2^k + 2^k = 2^{k+1}$ (because $2^k \geq 1$). Chaining, $k+1 \leq 2^k < 2^{k+1}$, so $k+1 < 2^{k+1}$. ✓

Problem 15 — [challenge] Spot the flaw (all horses). The hole is in the inductive step, and it shows up at the *very first link*, $k = 1 \rightarrow k = 2$. The step claims the “first k ” horses and the “last k ” horses *overlap*, so their colours chain together. For $k+1 = 2$ that means two groups of one horse each — horse #1 alone and horse #2 alone — and those groups share *no* horse, so there’s nothing to chain the colours through. The argument silently assumes $k \geq 2$ (where the groups really do overlap), so the step is *not* valid for every k ; the chain never gets going from 1 to 2, and the whole “proof” collapses. The base case was fine — one broken link in the step is all it takes. *Moral: an argument that looks like induction is worthless until the step is checked at every k , especially the first.*

Chapter 8 · Numbers That Shouldn't Exist

Things worth thinking about

1. **Multiplying $1 + i$ by i .** $(1 + i)i = i + i^2 = i - 1 = -1 + i$. The point $(1, 1)$ has moved to $(-1, 1)$ — a quarter turn counterclockwise. Yes, a 90° rotation.
2. **Why $e^{i\pi} = -1$.** After turning through angle π (half the circle), the walker stands at the leftmost point of the unit circle, whose address is $(\cos \pi, \sin \pi) = (-1, 0) = -1$. Half a lap lands on -1 .
3. **Roots of $x^2 + 1$.** $x^2 = -1$, so $x = \pm i$. The two roots are i and $-i$ — sitting on the imaginary axis at $(0, 1)$ and $(0, -1)$.
4. **The five extensions.** $\mathbb{N} \rightarrow \mathbb{Z}$ forced by *subtraction*; $\mathbb{Z} \rightarrow \mathbb{Q}$ by *division*; $\mathbb{Q} \rightarrow \mathbb{R}$ by *roots / filling gaps* (e.g. $\sqrt{2}$); $\mathbb{R} \rightarrow \mathbb{C}$ by *square roots of negatives* ($x^2 = -1$). And then it stops: over $\mathbb{C}$ *nothing* breaks — every polynomial already has all its roots (the Fundamental Theorem), roots and logs of complex numbers stay complex. That “completeness” is why the journey ends at $\mathbb{C}$.

Problems to work

Problem 1 — Arithmetic with i . (a) $(2 + 3i) + (4 - i) = \mathbf{6 + 2i}$. (b) $(2 + 3i) - (4 - i) = \mathbf{-2 + 4i}$. (c) $i^3 = i^2 \cdot i = \mathbf{-i}$. (d) $i^4 = (i^2)^2 = \mathbf{1}$ — so powers of i cycle $i, -1, -i, 1, \dots$ (e) $(1 + i)(1 - i) = 1 - i^2 = 1 + 1 = \mathbf{2}$ (purely real).

Problem 2 — A quadratic with no real roots. $x^2 + 4x + 13 = 0$: discriminant $16 - 52 = -36$, so $x = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = \mathbf{-2 \pm 3i}$. The roots $-2 + 3i$ and $-2 - 3i$ are a conjugate pair — same real part, opposite imaginary parts.

Problem 3 — Multiplication is rotation. $2 \xrightarrow{\times i} 2i \xrightarrow{\times i} -2 \xrightarrow{\times i} -2i \xrightarrow{\times i} 2$. Four quarter-turns carry 2 counterclockwise around the circle of radius 2 and back home.

Problem 4 — Use Euler's formula. $e^{i\pi/2} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = 0 + i \cdot 1 = \mathbf{i}$. It sits at $(0, 1)$ — straight up — exactly a quarter turn, matching the rotation picture.

Problem 5 — De Moivre at work. $(\cos 30^\circ + i \sin 30^\circ)^3 = \cos(3 \cdot 30^\circ) + i \sin(90^\circ) = \cos 90^\circ + i \sin 90^\circ = \mathbf{i}$. (Just triple the angle.)

Problem 6 — Teach it (the whole journey). *A teaching task — the goal is what happens in your head while you explain. A strong version hits: each*

new kind of number was *invented* to repair a broken operation — negatives for subtraction, fractions for division, irrationals for roots, and now i for $\sqrt{-1}$. None was “found in nature” (you’ve never held -2 apples either). i earns its keep doubly: it fixes the broken square root *and* turns out to *be* rotation. The keeper: *i is no more imaginary than -2 — both are inventions that make the mathematics work.*

Problem 7 — Find the roots of unity. $x^4 = 1$ with $x = \cos \theta + i \sin \theta$ gives, by De Moivre, $\cos 4\theta + i \sin 4\theta = 1$, so $4\theta = 0, 2\pi, 4\pi, 6\pi$, i.e. $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. The four roots are **1, i , -1 , $-i$** — the corners of a **square** on the unit circle. This matches the algebra $x^4 - 1 = (x^2 - 1)(x^2 + 1) = 0$, giving $\pm 1, \pm i$.

Problem 8 — [challenge] The conjugate trick. (a) $(a + bi)(a - bi) = a^2 - (bi)^2 = a^2 - b^2 i^2 = a^2 + b^2$ — always *real* (and ≥ 0). (b) Multiply top and bottom of $\frac{1}{2+i}$ by the conjugate $2 - i$:

$$\frac{1}{2+i} = \frac{2-i}{(2+i)(2-i)} = \frac{2-i}{4+1} = \frac{2-i}{5} = \frac{2}{5} - \frac{1}{5}i.$$

Multiplying by the conjugate makes the denominator real — that’s how you divide complex numbers.

Problem 9 — [challenge] A square root of i . i sits at distance 1, angle 90° . A square root should sit at distance 1 and *half* the angle, so squaring (which doubles the angle) returns to 90° . (a) Half of 90° is **45°** . (b) $z = \cos 45^\circ + i \sin 45^\circ = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$. (c) Check:

$$z^2 = \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)^2 = \frac{1}{2} (1 + i)^2 = \frac{1}{2} (1 + 2i - 1) = \frac{1}{2} (2i) = i. \quad \checkmark$$

(So i does have a square root — and $-z$ is the other one. In polar form, taking a root is just *dividing the angle*.)

Problem 10 — [challenge] Why exactly n roots. $x^6 = 1$ with De Moivre needs $6\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi$, so

$$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

Six equally spaced points — a regular **hexagon** on the unit circle. (c) Try one more, $6\theta = 12\pi$: that’s $\theta = 2\pi$, a full turn back to the $\theta = 0$ root you already have. The circle has room for exactly 6 evenly spaced angles before they start repeating — which is the geometric reason “degree n ” means “exactly n roots.”